Ontology-supported Integration of Gene Expression Data from Heterogeneous Subjects in KPMP and HuBMAP Resources for Systematic Kidney Disease Analysis

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Abstract

Precision medicine requires integrating diverse biomedical data. The NIH-funded Kidney Precision Medicine Project (KPMP, kpmp.org) collects clinical, pathological, and molecular data to advance personalized kidney disease treatment and tissue mapping. The Human Biomolecular Atlas Program project (HuBMAP, portal.hubmapconsortium.org) collects reference kidney cell and biomarker-related data. We developed an ontology-based platform to model and integrate this heterogeneous data, previously published and awarded at AMIA 2024 (Ref. 1). In this study, we expanded the platform to integrate both KPMP and HuBMAP gene expression and associated human subject datasets, identifying 48 variables (23 KPMP-specific, 13 HuBMAP-specific, and 12 shared). Gene expression patterns on known biomarkers (e.g., SPP1 and NBAS) and potentially new biomarkers (e.g., PDE4D) were identified in AKI (Acute Kidney Injury) and CKD (Chronic Kidney Disease) patients by comparing the normal control human subject populations in KPMP and HuBMAP.

Goal and Workflow

We previously used ontology-based methods to analyze KPMP data and identified gene markers linked to AKI and CKD (Ref. 1). By integrating KPMP and HuBMAP data into a unified system, we enable cross-database analysis of gene expression, phenotypes, and clinical variables, advancing precision kidney medicine.

Fig. 1 laid out the workflow of our study. Specifically, the raw data from HuBMAP Left Kidney (LK), HuBMAP Right Kidney (RK), and KPMP Single Nucleus (SN) sources are first collected. Observational and gene variables are extracted, and shared variables are used to integrate the datasets. The combined data is then quality-controlled and normalized to ensure consistency. After preprocessing, analytical techniques are applied to generate biologically meaningful insights (**Fig. 1**).

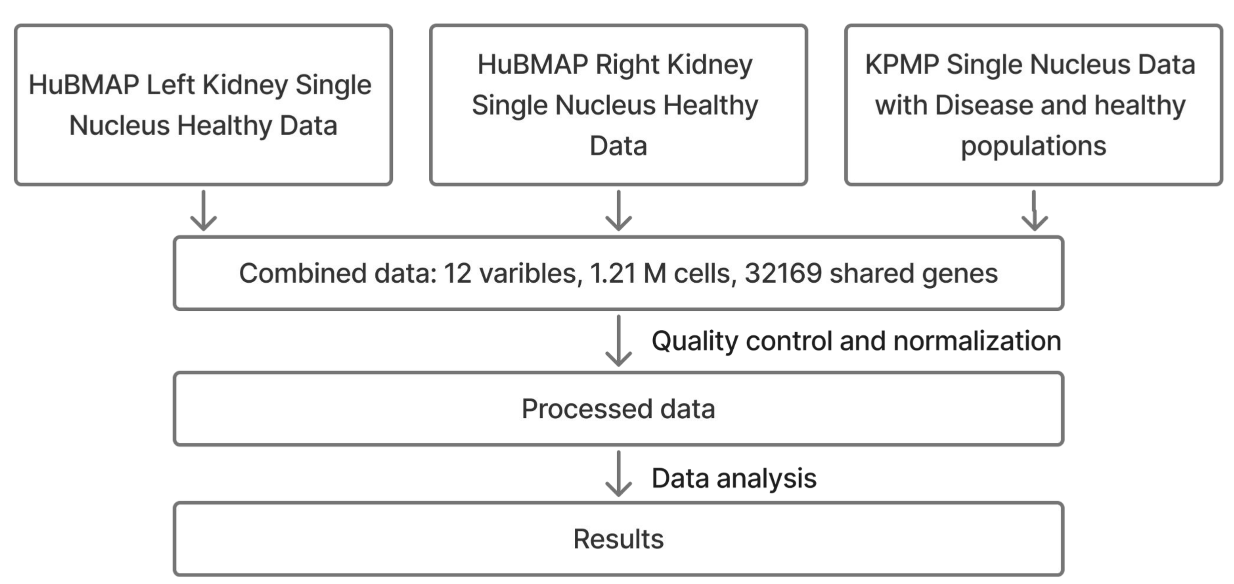
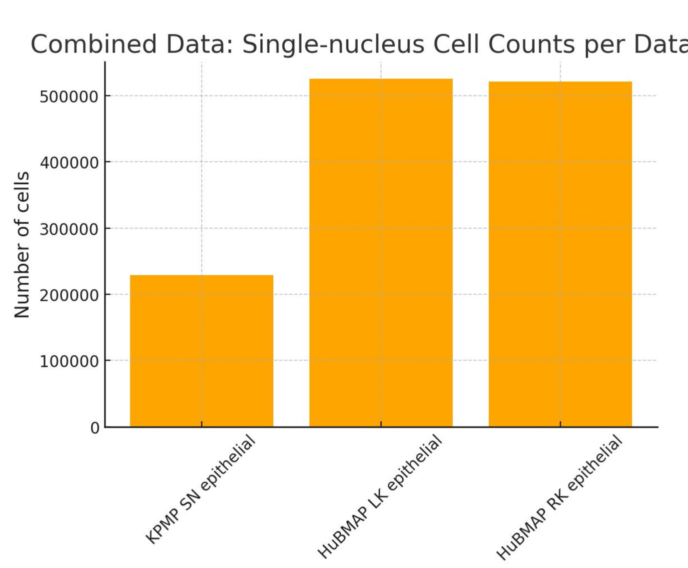


Fig. 1. Workflow for integrating and analyzing HuBMAP and KPMP single-nucleus RNA-seq data.

Fig. 2 illustrates the cell counts of kidney epithelial cell populations in both KPMP and HuBMAP datasets, which form a basis for our further integrative research.



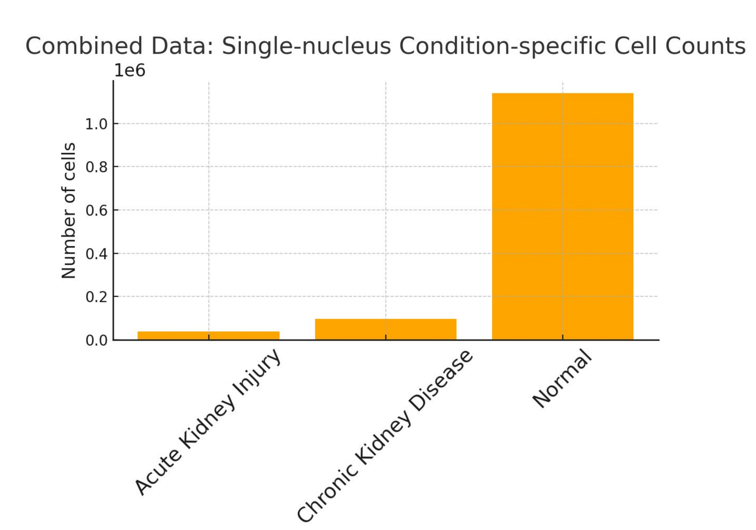
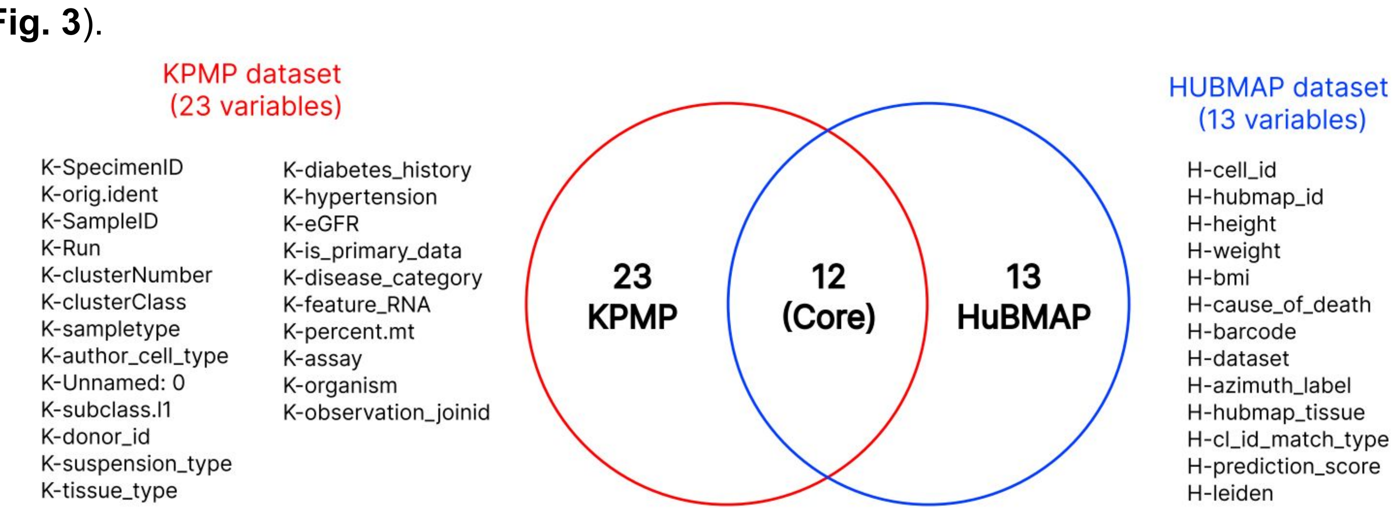


Fig. 2. Cell counts in kidney epithelial cell populations **(a)** Cell counts across KPMP and HuBMAP datasets. **(b)** Condition-specific epithelial cell counts in the combined dataset.

Data Integration

By integrating over 1.27 million single-nucleus kidney epithelial cell from KPMP and HuBMAP (covering 32,169 shared genes), we created a harmonized atlas cohort in which 38,300 AKI, 96,502 CKD, and 1,140,374 normal; 228,862 KPMP, 524,911 HuBMAP LK, and 521,403 HuBMAP RK cells could be directly compared (**Fig. 2**).

Furthermore, 23 unique variables from KPMP side, 13 unique variables from HuBMAP in both right kidneys (RK) and left kidneys (LK), and 10 observational homogeneous variables and 2 genetic homogeneous variables were found (**Fig. 3**).



Core
(shared by two datasets: 10 obs variables + 2 vars variables)

1. dataset_id ("LibraryID" or "library_id" in KPMP, "uuid" in HUBMAP)
2. cell_id ("*" in KPMP as hidden value, "cell_id" in HUBMAP)
3. as_id ("tissue_ontology_term_id" in KPMP, "UBERON:0002113" in HUBMAP as hidden value)
4. cl_id ("cell_type_ontology_term_id" in KPMP, "predicted_CLID" in HUBMAP)
5. cl_label ("cell_type" in KPMP, "predicted_label" in HUBMAP)
6. gene_count ("nCount_RNA" in KPMP, "n_counts" in HUBMAP)
7. age ("Age_binned" in KPMP, "age" in HUBMAP)
8. sex ("sex" in KPMP, "sex" in HUBMAP)
9. race ("self_reported_ethnicity" in KPMP, "race" in HUBMAP)
10. disease ("disease" in KPMP, "normal" in HUBMAP as hidden value)
11. gene_ID (First ID column for both KPMP and HUBMAP)
12. gene_name ("feature_name" in KPMP, "hugo_symbol" in HUBMAP)

Figure 3. Overlap of metadata variables between KPMP and HuBMAP datasets.

Harmonizing cell-type annotations between KPMP and HuBMAP using the Cell Ontology (CL) enables alignment across different levels of granularity. For example, KPMP labels “epithelial cell of proximal tubule,” while HuBMAP specifies its segmental subtypes, all children of the same CL term. Similarly, HuBMAP’s early and late distal tubule cells map to KPMP’s broader “distal convoluted tubule” label. These hierarchical relationships allow collapsing or expanding annotations for integrated analysis.To further explore how variables correspond between the KPMP and HuBMAP datasets, we developed an ontological framework that illustrates the relationships among each of these variables (**Fig. 4**).

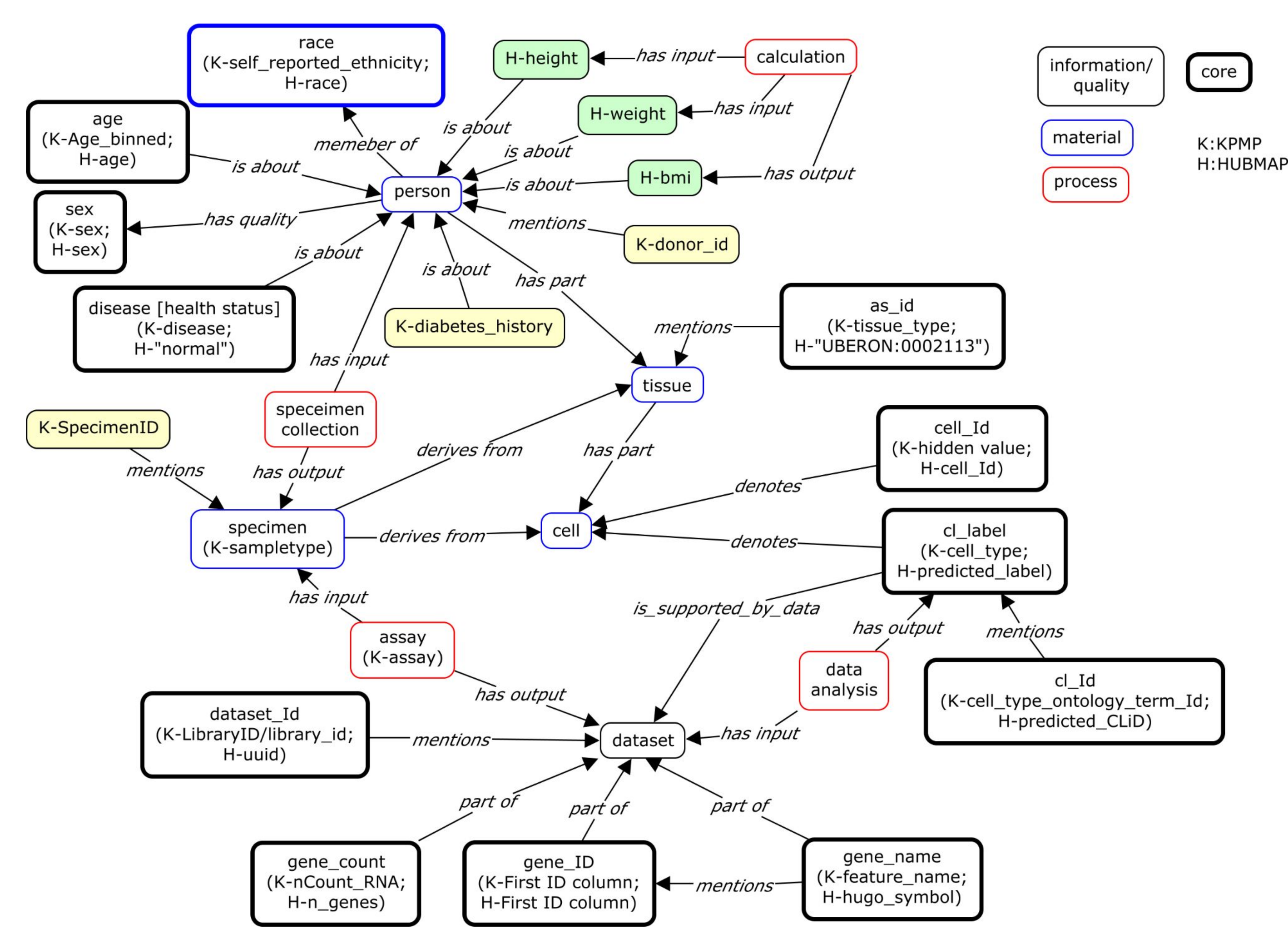


Fig. 4. Ontological framework illustrating the relationships among variables in KPMP & HuBMAP datasets.

Profiling Kidney Disease Biomarkers

We have analyzed the gene expression profiles of those AKI and CKD biomarkers collected in our previous KPMP research using integrative KPMP and HuMBAP datasets. Our general results were in agreement with previous findings that were discovered on only the KPMP dataset (Ref. 1) (**Fig. 5**). Current research showed that the biomarker genes such as SPP1 is highly expressed in AKI epithelial cell populations, confirming its potential role in kidney epithelial cell damage (Ref. 1). Furthermore, we also made several new findings, for example, biomarker gene NBAS (NBAS Subunit Of NRZ Tethering Complex) overexpressed in many kidney epithelial cell types in the AKI population compared to the normal control (**Fig. 5**), suggesting its potential role in AKI pathogenesis.

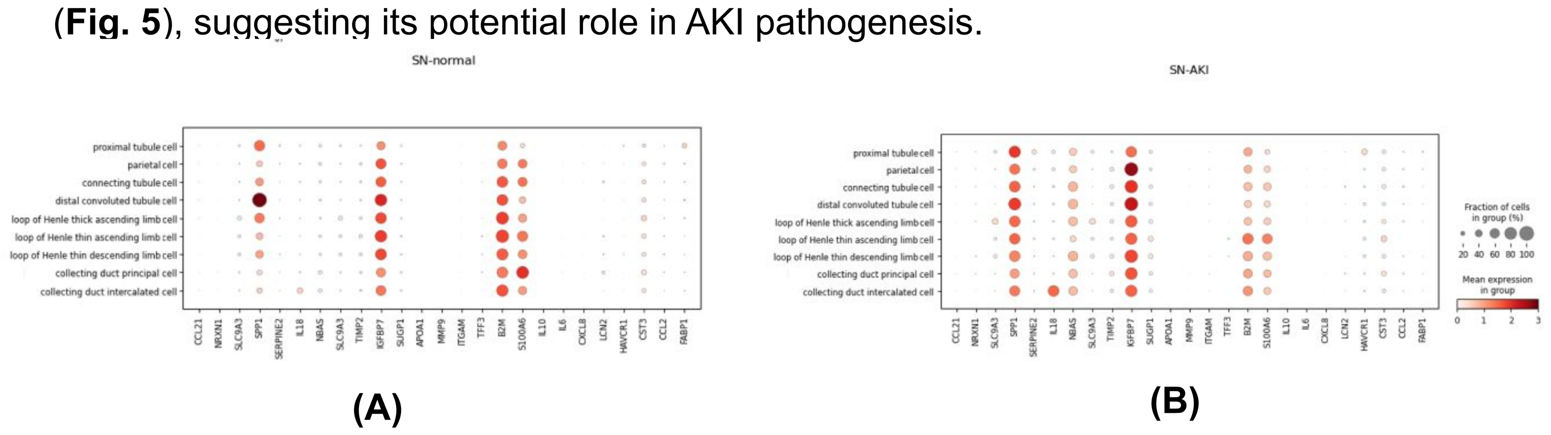


Fig. 5. Dot plots of known gene markers in cell types in normal (A) and AKI (B) conditions.

Wilcoxon Rank-Sum Test for Differential Gene Expression Analysis

By analyzing all the available genes (instead of biomarkers), our Wilcoxon rank-sum test found many interesting results. Fig. 6 showed the top 20 genes identified. PDE4D (phosphodiesterase 4D) is ranked top 1 among the top 20 differentially expressed genes in AKI (A) and CKD (B) populations compared over the healthy control (**Fig. 6**). The analysis of the top 20 gene lists found five genes unique to AKI, and five unique to CKD.

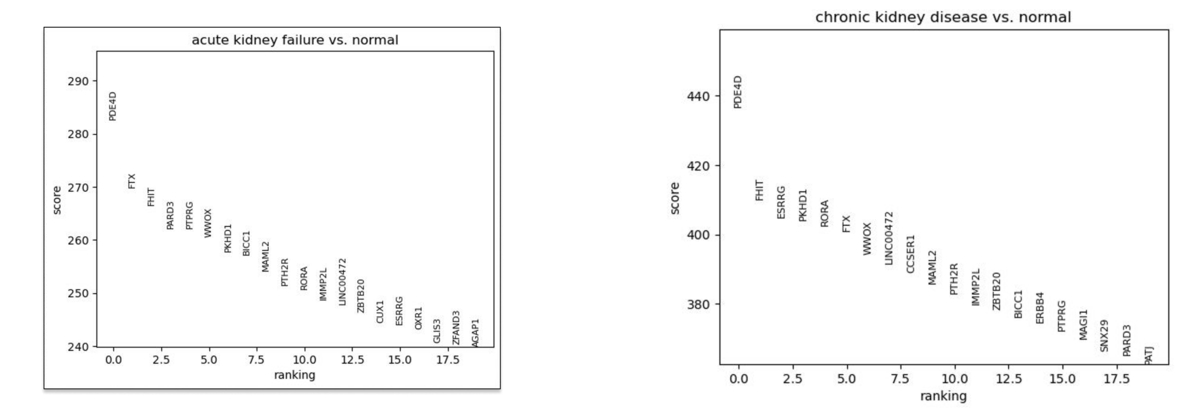


Fig. 6. Top differentially expressed genes in kidney epithelial cells from diseased human subjects when compared with the normal control. (a) AKI vs. normal; (b) CKD vs. normal. Differential gene expression analysis was conducted using the Wilcoxon rank-sum test after preprocessing and normalization ($p < 0.05$). Genes are ranked by significance (x-axis), and the test statistic (“score”) is shown on the y-axis. Higher scores reflect elevated expression in disease conditions.

Conclusion

Our ontology-guided integration of KPMP and HuBMAP single-nucleus kidney epithelial cell datasets proved to be successful, enabling the robust profiling of known disease-associated markers (e..g, SPP1 and NBAS) and identification of potentially new biomarkers such as PDE4D. The ontological modeling provides a basis on the integration of heterogeneous datasets at the metadata level. More analysis is being conducted to further improve our ontological modeling and data analysis.

References

1. He, Yongqun Oliver, et al. "Ontology-based modeling, integration, and analysis of heterogeneous clinical, pathological, and molecular kidney data for precision medicine." *AMIA Annual Symposium Proceedings*. Vol. 2024. 2025.

Acknowledgements

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